PARATHYROID DISEASE

AT A GLANCE

- The most common cause of hypercalcemia in outpatients is hyperparathyroidism.
- The most common cause of hypercalcemia in hospitalized patients is malignancy.
- The most common causes of hypocalcemia are vitamin D deficiency and surgically induced hypoparathyroidism.
- Calciphylaxis is most commonly seen in the setting of secondary hyperparathyroidism of renal failure.

Epidemiology

The annual incidence of primary hyperparathyroidism has decreased over the past two decades to approximately 20 cases per 100,000, possibly due to reduced use of ionizing radiation to the neck—a known risk factor for the disease. Most cases occur in individuals over 45 years of age, with a female-to-male ratio of about 2:1. In outpatient settings, parathyroid disease accounts for nearly all cases of hypercalcemia. Among hospitalized patients, parathyroid disease causes about one-third of hypercalcemia cases, with malignancy responsible for the majority of the remaining cases. Together, hyperparathyroidism and malignancy account for over 90% of all hypercalcemia cases.

Etiology and Pathogenesis

Parathyroid hormone (PTH) is secreted by the parathyroid glands in response to low serum calcium levels and is a key regulator of calcium homeostasis in the body. PTH increases serum calcium by stimulating bone resorption, enhancing renal calcium reabsorption, and promoting the activation of vitamin D, which in turn increases intestinal calcium absorption. Dysregulation of PTH secretion

—whether due to adenoma, hyperplasia, or carcinoma of the parathyroid glands —leads to primary hyperparathyroidism and subsequent hypercalcemia. Secondary hyperparathyroidism occurs in response to chronic hypocalcemia, most commonly in the context of chronic kidney disease. Tertiary hyperparathyroidism refers to autonomous PTH secretion following prolonged stimulation, often after kidney transplantation.